

VISTRA - Detailed Solution Document

PS 26126 | Vision Based Autonomous Navigation for Unmanned Ground Vehicle for Outdoor Environment

Evidence policy: PS-specific VISTRA content is from the supplied blueprint (B1). External technical and dataset facts are tied to the numbered source register. Project performance numbers are not claimed until measured.

1. Executive summary

VISTRA is a camera-first autonomy stack for an outdoor unmanned ground vehicle (UGV). The supplied blueprint defines its thesis as carrying uncertainty from image perception into the risk map, into planning, and finally into motion behavior. The goal is to make safety decisions explicit, measurable and recoverable rather than to add the largest feature set.

Camera	->	Perception	->	BEV risk	->	VO + EKF	->	D* Lite	->	DWA	->	TTC	->	Wheels
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The dashboard is part of the evidence chain: it shows camera segmentation, BEV risk, pose/path, Navigation Confidence Score (NCS), TTC, state, wheel commands, decision logs, fault injection and baseline comparison.

Evidence: B1 Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF)

2. PS interpretation and traceability

The supplied blueprint interprets PS 26126 as requiring lightweight perception, visual odometry/SLAM, a planner producing wheel/motor commands, and collision-free Point-A-to-Point-B navigation across outdoor scenarios.

Requirement	VISTRA module	Evidence
Lightweight perception	SegFormer-B0 + ONNX	Per-class IoU, FPS, latency
Visual localization	Stereo VO/SLAM + EKF	ATE/drift, pose trace, covariance
Planning	D* Lite + DWA	Path, replans, route behavior
Wheel commands	(v,w) mapper	Command log + wheel panel
Collision-free A to B	Full stack	Seeded scenario results

This is the target evidence plan from B1, not a claim that the implementation already satisfies every requirement.

Evidence: B1 Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF)

3. Architecture and feedback loops

The architecture separates perception, localization, planning and safety but lets confidence propagate between them. Pose covariance inflates map cost. NCS caps speed. Localization failure invokes recovery. Planning failure escalates to replan, rotate-search and safe stop.

Input quality		->	Segmentation		->	Ground projection		->	Risk map		->	Confidence	
Stereo VO	->	Pose/covariance	->	D* Lite	->	DWA	->	TTC	->	Governor	->	Wheel mapper	

A hardware abstraction layer keeps camera, vehicle and sensor implementations out of the navigation core. The blueprint lists simulation, recorded-video, webcam and optional physical-vehicle modes.

Evidence: B1 Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF)

4. Perception and traversability

VISTRA uses semantic terrain segmentation and dynamic-object tracking. The six proposed classes are smooth-drivable, rough-drivable, vegetation/non-traversable, hard obstacle, water/ditch/negative, and background/sky.

Quantity	Definition in blueprint
Traversability	$T = \sum(p_k * t_k)$
Uncertainty	$U = \text{normalized entropy}(p)$
Cell risk	$R = (1-T) + \lambda_u U + \lambda_g G + \lambda_{prox} P$
Example BEV	20 m x 20 m at 0.2 m resolution (100 x 100 cells)
Risk bands	Green <0.25; yellow <0.5; orange <0.75; red >=0.75/lethal

The design is free-space-first: confidently drivable space is preferred while unknown or uncertain terrain becomes costly instead of silently free. The blueprint explicitly notes that ditches can be hard to recall from appearance alone.

Limitation: IPM assumes locally flat ground and can distort on slopes. Proposed mitigation: IMU pitch/roll compensation and lower far-range weighting.

Evidence: B1 Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF)

5. Localization without GPS

The primary path is stereo visual odometry with loop-closure relocalization, using RTAB-Map or ORB-SLAM3, fused with simulated IMU and wheel odometry in an EKF. A fallback uses OpenCV features, PnP and RANSAC.

ORB-SLAM3 is a research-backed candidate because its paper describes visual and visual-inertial SLAM with monocular, stereo and RGB-D cameras and a multi-map system. This establishes capability fit, not VISTRA performance.

Confidence signal	Purpose
Tracked feature count	Detect feature starvation
RANSAC inlier ratio	Check geometric match quality
Reprojection error	Check pose consistency
EKF covariance	Measure growing state uncertainty
Input quality	Blur, brightness, feature distribution

Recovery sequence from B1: confidence falls -> slow -> stop -> relocalize -> retrace toward the last good keyframe -> resume after recovery. If recovery fails, temporary dead reckoning uses IMU/wheel data while covariance grows.

Evidence: B1 Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF) | S6 ORB-SLAM3 paper

6. Planning and safety

Global planning uses D* Lite on a coarse cost grid. D* Lite is an incremental heuristic search method designed to reuse prior search information as the navigation problem changes. Local planning uses the Dynamic Window Approach (DWA), which searches reachable velocity commands while incorporating robot dynamics.

Evidence: **S7** D* Lite - Carnegie Mellon Robotics Institute | **S8** Dynamic Window Approach - Carnegie Mellon Robotics Institute

The VISTRA safety layer forward-simulates the vehicle for 3 seconds at 0.1-second steps. Dynamic obstacles use a constant-velocity model. TTC is the first predicted step with footprint overlap.

Condition	Starting action in B1
$TTC > 3 \cdot \tau_{stop}$	Continue
$1.5 \cdot \tau_{stop} < TTC \leq 3 \cdot \tau_{stop}$	Slow and replan
$\tau_{stop} < TTC \leq 1.5 \cdot \tau_{stop}$	Hard brake and replan/evade
$TTC \leq \tau_{stop}$	Emergency stop

Stopping time is $\tau_{stop} = t_{latency} + v/a_{max}$. The latency term must be measured end-to-end. Planner output (v,w) is mapped to differential-drive commands using $v_{left} = v - wL/2$ and $v_{right} = v + wL/2$, then clamped to motor limits.

Evidence: **B1** Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF)

7. Confidence engine and recovery states

NCS is a weakest-link behavior gate built from perception confidence, localization confidence, path confidence, input quality and optional depth/geometry agreement.

NCS	State	Behavior
> 0.7	NORMAL	Normal speed cap
0.4 - 0.7	CAUTION	Speed cap down; wider inflation
0.2 - 0.4	RECOVER	Stop/creep; relocalize/re-observe
< 0.2	SAFE_STOP	Hold and alert

Hysteresis is used to prevent rapid state oscillation. Every state transition writes a reason to the decision log.

Fault	Detection	Response
Poor light	Brightness/entropy spike	Slow + inflate costs
Motion blur	Laplacian variance drop	Reduce speed
Camera obstruction	Variance/features collapse	Stop + flag
Featureless terrain	Low tracked features	Slow + dead reckon
Localization failure	Covariance/inlier collapse	Stop + relocalize/retrace
Input loss	Frame timeout	Stop after grace period
Planning failure	No path	Rotate-search/back up/replan

Evidence: B1 Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF)

8. Simulation, datasets and benchmark

Closed-loop navigation is planned in Gazebo Harmonic + ROS 2 Jazzy. The real-video track runs perception, mapping and confidence on outdoor footage and is described as open-loop. Current Gazebo documentation lists Harmonic as an LTS release with May 2029 EOL.

RUGD was created for unstructured outdoor driving and reports 7,453 pixel-wise labeled images. RELIS-3D's official repository reports 13,556 annotated LiDAR scans and 6,235 images, with a 20-class ontology and multimodal sensor data.

Source	Verified fact	Planned use
RUGD	7,453 labeled images	Outdoor semantic terrain research
RELIS-3D	13,556 LiDAR scans; 6,235 images; 20 classes	Off-road multimodal research
Synthetic simulation	Controlled labels / ground truth	Closed-loop benchmark
Own data	About 300-500 frames planned	Held-out real-video validation

Evidence: **B1** Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF) | **S3** Gazebo Harmonic documentation | **S9** RUGD paper / official host | **S10** RELIS-3D official repository | **S11** RELIS-3D paper

9. Validation protocol and evidence

The benchmark is designed around reproducibility. Scenarios include normal terrain, difficult terrain, sudden obstacle, moving obstacle, poor lighting, localization degradation, perception failure, narrow path, dead end, multiple routes and unknown obstacle.

Protocol	20-30 seeded runs per scenario per system; report mean and confidence interval
Metrics	Success, collisions, path length, time, localization error, path deviation, replans, FPS, latency, NCS trace, recovery time
Baseline	YOLOv8 detection + same VO/SLAM + A* on binary occupancy grid + fixed-distance stop
Ablations	Remove uncertainty term; remove confidence governor; remove TTC/terrain safety

The blueprint explicitly says not to fabricate numbers. If the baseline performs better on a scenario, that result should be reported.

Evidence: B1 Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF)

10. Roadmap, risks and scope control

Phase	Weeks	Definition of done
Simulation + vehicle	1	Teleop vehicle; camera/IMU topics; seeded worlds
Perception	1-3	SegFormer-B0 ONNX; IoU/FPS measured
Traversability	2-3	BEV risk map + fusion
Localization	2-4	VO + EKF; ATE logged
Planning	3-5	D* Lite + DWA drives A to B
Safety	5-6	TTC tested on injected obstacles
Confidence/recovery	6-7	NCS gates speed; faults recover
Dashboard	3-8 parallel	All panels live
Benchmark	7-9	Baseline + ablations
Final demo	9-10	Three rehearsals including backup

Major risks in B1 are SLAM integration time, sim-to-real perception gap, weak ditch detection, live simulation failure, setup friction, excessive latency, scope creep and overfitting. Mitigations include time-boxing, a separate real-video track, uncertainty/geometry terms, rosbag/video backup, common environment setup and sequence-level splits.

Scope rule: freeze new features at the end of week 6; use the remaining period for testing, benchmarking and rehearsal.

Evidence: B1 Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF)

11. Demonstration script and judge answers

- Start without GPS; show camera feed and empty map.
- Show segmentation and BEV risk accumulation.
- Drive toward the goal with the global path visible.
- Inject an obstacle; show TTC/risk rise, speed reduction and replanning.
- Inject blur/dark fault; show NCS fall, RECOVER and relocalization.
- Reach goal; show collisions, replans and recovery-time summary.
- Play side-by-side baseline on the same scenario seed.

Common questions should be answered with the documented boundary: real-video perception is open-loop, closed-loop navigation is initially validated in simulation, ditches are a known limitation, and final threshold values must be tuned from seeded runs.

Evidence: B1 Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF)

Appendix - Trusted Source Register

B1 - Supplied VISTRA - SIH 2026 Solution Blueprint (reference PDF)

PS 26126 metadata, VISTRA design, formulas, roadmap, benchmark protocol, risks and limitations.

Supplied project source PDF.

S1 - Bharat Electronics Limited - Official Homepage

Official BEL organizational description.

<https://bel-india.in/homepage/>

S2 - ROS 2 Jazzy documentation

ROS 2 Jazzy documentation and packages.

<https://docs.ros.org/en/jazzy/>

S3 - Gazebo Harmonic documentation

Official Gazebo Harmonic documentation and release information.

<https://gazebosim.org/docs/harmonic/>

S4 - SegFormer paper

Original SegFormer architecture and efficiency rationale.

<https://arxiv.org/abs/2105.15203>

S5 - Hugging Face SegFormer documentation

Model configuration; MiT-B0 listed at 3.7M parameters.

https://huggingface.co/docs/transformers/main/model_doc/segformer

S6 - ORB-SLAM3 paper

Visual and visual-inertial SLAM, stereo support and multi-map behavior.

<https://arxiv.org/abs/2007.11898>

S7 - D* Lite - Carnegie Mellon Robotics Institute

Incremental heuristic replanning for unknown terrain.

<https://publications.ri.cmu.edu/d-lite>

S8 - Dynamic Window Approach - Carnegie Mellon Robotics Institute

Velocity-space local collision avoidance using robot dynamics.

<https://publications.ri.cmu.edu/the-dynamic-window-approach-to-collision-avoidance>

S9 - RUGD paper / official host

7,453 pixel-wise labeled images; unstructured outdoor driving challenges.

https://rugd.vision/pdfs/RUGD_IROS2019.pdf

S10 - RELLIS-3D official repository

13,556 LiDAR scans, 6,235 images, 20-class ontology and multimodal data.

<https://github.com/unmannedlab/RELLIS-3D>

S11 - RELLIS-3D paper

Off-road semantic scene understanding dataset motivation and benchmarks.

<https://arxiv.org/abs/2011.12954>

S12 - ONNX Runtime documentation

Cross-platform ML inference runtime.

<https://onnxruntime.ai/docs/>

S13 - CVAT documentation

Visual dataset annotation and management platform.

https://docs.cvat.ai/docs/getting_started/overview/

S14 - OpenCV documentation

Computer-vision implementation APIs.

<https://docs.opencv.org/>

S15 - FastAPI documentation

Python API framework used for dashboard integration.

<https://fastapi.tiangolo.com/>